

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Data Sheets for Electrical Equipments
Document Number	EA-BFPCL-GPMC-GGPL-FEED-ELE-DSH-002
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Data Sheets for Electrical Equipments

R01	08.08.22	Issued for Review	N. Kari	V. Ani	A.Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	08.08.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ELE-DSH-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

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				R01					
Client		Epic Atlantic Limited		Plant		FEED for Gas Gathering Pipelines			
Tag Nos		Hold		Location		MANIFOLD			
SERVICE: LV 400V Switchboard Tag no. SB01									
GENERAL REQUIREMENTS BY PURCHASER									
SYSTEM VOLTAGE		400 V		DEGREE OF PROTECTION		IP42 min			
VOLTAGE TOLERANCE		+/- 5%		CONSTRUCTION		Free standing			
FREQUENCY TOLERANCE		+/- 2%		ARRANGEMENT		Front access			
SYM. RMS FAULT CURRENT		50kA for 1 sec min		MOUNTING: INCOMER		Withdrawable			
ASYM. PEAK FAULT CURRENT		VTA		BUS TIE		NA			
SYSTEM EARTHING		Solid		FEEDERS		Withdrawable			
ATMOSPHERE		Tropical, Non-saliferous		BARRIER/PARTITION		To IEC 60439 Form 4B			
AMBIENT TEMPERATURE		25 Deg C		ARC CONTAINMENT		To IEC 60641			
RELATIVE HUMIDITY		100%		ENTRIES		Cable -bottom			
ALTITUDE ABOVE SEA LEVEL		<1000m		EXTERNAL PAINT COLOUR		Supplier's standard			
HAZ AREA CLASSIFICATION		Non-Hazardous		SPARE CUBICLES		Equipped			
LOCATION		Indoors, ventilated Room		ANTI-COND. HEATERS		230V, Single Phase			
PROT'N & METERING SLD		VTA							
ELECTRICAL SINGLE LINE DIAGRAM		EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-015							
Low Voltage Switchgear and Control Gear Assemblies (Amendments/supplements to IEC 60439-1).									
GENERAL DATA BY VENDOR									
SW'BD RATED VOLTAGE		400 TPN+E		DESIGN STANDARDS		IEC 61439			
BUSBAR ARRANGEMENT		Single		S/C TESTING AUTHORITY					
BUSBAR RATED CURRENT		120 A @ °C		TEST CERT Nos.: BUSBAR					
BUSBAR NEUTRAL RATING		VTA A @ °C		CIRCUIT BREAKERS					
BUSBAR FAULT RATING		15 kA for 1 sec.							
EARTH BAR CROSS SECTION		VTA mm ²		FLOOR FRAME REQUIRED		YES /NO mm			
MIN MAINTENANCE ACCESS		FRONT VTA VTA REAR		FOUNDATION PLAN DWG No.					
				G.A. DRAWING No.					
DIMENSIONS: OVERALL		VTA mm LxWxH		PANEL WEIGHTS: OVERALL		VTA		kg	
INCOMER		VTA mm LxWxH		INCOMER		VTA		kg	
BUS SECTION		VTA mm LxWxH		BUS SECTION		VTA		kg	
OUTGOING TIER		VTA mm LxWxH		OUTGOING TIER		VTA		kg	
SHIPPING SECTIONS				HEAT EMISSION					
IDENTIFIER		DIMENSIONS		WEIGHT				kW @ 100% LOAD	
				kg				kW @ 75% LOAD	
				kg				kW @ 50% LOAD	
				kg					
NOTE:									
1. VTA : Vendor To Advice									
2. NA: Not Applicable									

				Revision Index					
				Rev. No.					
				R01					
Client	Epic Atlantic Limited			Plant	FEED for Gas Gathering Pipelines				
Tag Nos	Hold			Location	MANIFOLD				
Datasheet - Distribution Boards									
1	GENERAL REQUIREMENTS								
1.1	Installation Site			Gas Gathering Manifold Stations					
1.2	Equipment Specification			EA-BFPCL-GPMC-GGPL-FEED-ELE-SPC-002 - Specification for Electrical Equipment					
1.3	Applicable Local Standards			IEC 60947					
1.4	Reference Drawings/Documents			EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-015 - LV one line diagram					
1.5	Erection And Commissioning By Supplier			Yes					
1.6	Language - On Documents			English					
1.7	- On Equipment			English					
1.8	Paint Colour			BS 4800 Grey Shade 10-A-3					
1.9	No. Of Unit			3					
1.1	Altitude			Less Than 1000m Above Sea Level					
2	SERVICE CONDITIONS								
2.1	Location			Indoor					
2.2	External Ambient Temperature			Maximum		50°C	Minimum		-3°C
2.3	External Ambient Relative Humidity			Maximum		100%			
2.4	Internal Ambient Temperature			Maximum		50°C	Normal		24°C Minimum 5°C
2.5	Internal Ambient Relative Humidity			Maximum		100%			
2.6	Normal			Air Conditioned					
2.7	Emergency			**					
2.8	Unusual Exposure To			Excessive Dust - Yes		Excessive Salt - Yes			
3	TECHNICAL DATA								
3.1	Distribution Board Type			Fixed Type, Front Operated					
3.2	System Voltage And Frequency			400 V, 50 Hz, 3Ph , 4 WIRE					
3.3	Supply Voltage Tolerance			+/- 5%					
3.4	Supply Frequency Tolerance			+/- 2%					
3.5	Neutral Earthing			SOLID					
3.6	Prospective Short Circuit Current			25 kA Symmetrical FOR 1 SEC.					
3.7	Anti Condensation Heating			230 V AC (Internally Derived – Refer Note-8)					
3.8	Cable Entry			Bottom					
3.9	Cable Type			Cu/XLPE/SWA/PVC					
3.1	Cable Gland			Double Compression Type (By Purchaser)					
4	MAIN BUS-BAR SYSTEM								
4.1	Bus Bar Arrangement			As Per Respective SLD mentioned In Item No. 1.5 of this datasheet					
4.2	Phase Insulation			Fully Insulated, Flame retardant, Non-Hygroscopic, Non tracking					
4.3	Ingress Protection Class			IP 42					
4.4	Short Time Withstand Current			25 kA For 1 second					
4.5	Nominal Voltage			400/230VAC					
4.6	Main Bus Bar Continuous Current Rating At Ambient Conditions			As Per Respective SLD mentioned In Item No. 1.5 of this datasheet					
4.7	Bus-Bar Material			Hard drawn high conductivity copper					
5	DISTRIBUTION BOARD CONSTRUCTION (GENERAL)								
5.1	Area Classification			Safe Area					
5.2	Ingress Protection			IP 42					
5.3	Enclosure Protection			EX 'd' Flameproof / Industrial Type					
5.4	Incomer			MCCB, 4 Pole (Ratings shall be as per respective SLD mentioned In Item No. 1.4 of this datasheet)					
5.5	Outgoing Feeders			4 Pole / 3 Pole / 2 Pole MCB's with ELCB (For Feeder details refer respective SLD mentioned In Item No. 1.4 of this datasheet)					
5.6	Mounting			Floor Mounted					
5.7	Access			Single Front					
5.8	Incomer Cable Size			As per respective SLD mentioned In Item No. 1.4 of this datasheet					
5.9	Earth Cable Size			2 Nos 1Cx35mm2 Green/Yellow					
5.10	Earth Busbar Size			25 x 6 mm2 Copper					
5.11	Other Requirements			Separate terminal block for outgoing cables with klippon or similar type pressure terminals					
5.12	Spare Feeders			25%					
5.13	Copy Of Type Test Certificate			Required					
5.14	Routine Test Certificate			Required					
5.15	Spare Terminals			25%					

				Revision Index					
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				R01					
Client	Epic Atlantic Limited			Plant	FEED for Gas Gathering Pipelines				
Tag Nos	Hold			Location	MANIFOLD STATION				
Datasheet - Distribution Boards									
6	DATA TO BE SUBMITTED BY VENDOR								
6.1	Manufacturer / Model No.			**					
6.2	Work Address			*					
6.3	Work Order No.			*					
6.4	Overall Equipment Dimensions			**					
6.5	Equipment Total Weight			**					
6.6	Painting Details			**					
6.7	Voltage Transformers			**					
6.8	Current Transformers			**					
6.9	Compliance With Specification/Data Sheet			**					
6.10	Deviation Submitted			**					
6.11	Busbar Rating			*					
6.12	Type Test Certificate Submitted			**					
6.13	MCCBs Prospective Rms Fault Current			**					
6.14	Peak Fault Making Current			**					
6.15	Busbar Size & Rating At The Specified Ambient Conditions (After Derating)			**					
6.16	Enclosure Material			Sheet Steel					
6.17	Maker's Name For MCB's , MCCB's And RCD's			**					
6.18	Reference Drawing			**					
Notes:									
1) ** data to be furnished by Supplier/Vendor/Manufacturer as per Project specification/International standard requirements.									
2) * data to be furnished by Contractor(later)									
3) For Detail Environmental Site and Climatic Conditions -									
4) Under Normal condition system Voltage $\pm 5\%$ and Frequency $\pm 2\%$, Under Transient condition Voltage $\pm 10\%$ and Frequency $\pm 5\%$									
5) Tests shall be carried out as per OEM standards in accordance with IEC standards, all certificates shall be available at the quotation stage									
6) All ratings shall be for ambient temperature of 50 deg. C									
7) Incomer shall be MCCB type with adjustable ambient temperature compensated over current (thermal and short circuit) protection. Incomer shall be pad lockable in off position.									
8) The distribution board shall be provided with incomer line VT, voltmeter and selector switch for reading phase / line voltage and also with red, yellow and blue indication lamps on the bus side.									
9) The distribution board shall be provided with 3nos. Of incomer line CT, ammeter and selector switch for reading phase / line current.									
10) Incomer & outgoing circuits, MCCB/MCBS shall be located in a single compartment. MCCB/MCBS shall be mounted on inner plate with separate compartment door outside.									
11) Lighting feeders shall be provided with earth leakage protection with 300 mA sensitivity.									
12) Socket outlet feeders shall be provided with earth leakage protection with 30 mA sensitivity.									
13) Thermostatically controlled anti-condensation heater operating on internally derived 230 V single phase supply complete with MCB & neutral link shall be provided. Space heater 'on' indication lamp shall be provided.									
14) Photoelectric cell with enclosure (IP-55) shall be provided. Auto/manual selector switch with override test facility shall also be provided.									
15) Cable glands shall be supplied by purchaser. Cable gland entries shall be iso metric thread. Entry sizes shall be finalized during drawing approval stage									
16) In case of conflict with the requirements specified in drawings, data sheet and standard, vendor shall follow most stringent requirement									
Reference Drawings/Documents									
EA-BFPCL-GPMC-GGPL-FEED-ELE-SPC-002- Specification for Electrical Equipment									
EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-015 - Single Line Diagram									
EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-001 - Electrical Load List									

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